www.tntextbooks.in

based on three references namely hydrogen, oxygen and chlorine, because we can not conceive of reactions involving only with those three references. Therefore, a more useful expression used to calculate gram equivalent mass is given below.

$$\text{Gram equivalent mass} = \frac{\text{Molar mass (g mol}^{-1})}{\text{Equivalence factor (eq mol}^{-1})}$$

On the basis of the above expression, let us classify chemical entities and find out the formula for calculating equivalent mass in the table below.

1.5.1 Equivalent Mass of Acids, Bases, Salts, Oxidising Agents and Reducing Agents

Chemical entity	Equivalent Factor(n)	Formula for calculating equivalent mass (E)	Example
Acids	basicity (no. of moles of ionisable H^+ ions present in 1 mole of the acid)	$E = \frac{\text{Molar mass of the acid}}{\text{Basicity of the acid}}$	H_2SO_4 basicity = 2 eq mol^{-1} Molar mass of H_2SO_4 = $(2 \times 1) + (1 \times 32) + (4 \times 16)$ = 98 g mol^{-1} Gram equivalent of H_2SO_4 = $\frac{98}{2}$ = 49 g eq^{-1}
Bases	Acidity (no. of moles of ionisable OH^- ion present in 1 mole of the base)	$E = \frac{\text{Molar mass of the base}}{\text{Acidity of the base}}$	KOH acidity = 1 eq mol^{-1} Molar mass of KOH = $(1 \times 39)+(1 \times 16)+(1 \times 1)$ = 56 g mol^{-1} Gram equivalent mass of KOH = $\frac{56}{1}$ = 56 g eq^{-1}
Oxidising agent (or) reducing agent	No. of moles of electrons gained (or) lost by one mole of the reagent during redox reaction	$E = \frac{\text{Molar mass of the oxidising (or) reducing agent}}{\text{no. of moles of electrons gained by one mole of the oxidising (or) reducing agent}}$	$KMnO_4$ is an oxidising agent, Molar mass of $KMnO_4$ = $(1 \times 39) +(1 \times 55) + (4 \times 16)$ = 158 g mol^{-1} In acid medium permanganate is reduced during oxidation and is given by the following equation, $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ $\therefore n = 5$ eq mol^{-1}. $\therefore$ equivalent mass of $KMnO_4 = \frac{158}{5} = 31.6$ g eq^{-1}.

Mole concept requires a balanced chemical reaction to find out the amount of reactants involved in the chemical reaction while gram equivalent concept does not